Math 5050 – Special Topics: Differential Geometry– Fall 2025
w/Professor Berchenko-Kogan

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Section 22: Bounded Manifolds – October 22, 2025

Key Points

Remark 0.0.1. Charts map boundary points to boundary points. That is, (U, ϕ) is a chart on M then $p \in \partial M$ implies $\phi(p) \in \partial\phi(U)$.

Remark 0.0.2. Topological boundaries/properties are taken from the *ambient* space (as if we are ignoring the manifold that it contains). Manifold properties are always measured from "within", that is, proved via the charts within the atlas that fully defines the Manifold.

22.1 Topological boundary versus manifold boundary

Let M be the subset $[0, 1) \cup \{2\}$ of the real line. Find its topological boundary $\mathrm{bd}(M)$ and its manifold boundary ∂M.

Since $[0, 1)$ is a one dimensional manifold its the boundary must be $\mathrm{bd}([0, 1)) = \{0\}$. The singleton set $\{2\}$ is a zero-dimensional manifold thus has no boundary.

22.2 Topological boundary of an intersection

Let A and B be two subsets of a topological space S. Prove that

$$\mathrm{bd}(A \cap B) \subset \mathrm{bd}(A) \cup \mathrm{bd}(B).$$

Consider a proof by contrapositive, that is, $x \notin \mathrm{bd}(A \cap B) \implies x \notin \mathrm{bd}(A) \cup \mathrm{bd}(B)$. Note that $\mathrm{bd}(A)^c = A^o \cup S \backslash \overline{A}$ then,

$$\begin{aligned}
&x \in (\mathrm{bd}(A) \cup \mathrm{bd}(B))^c \\
&x \in \mathrm{bd}(A)^c \cap \mathrm{bd}(B)^c \\
&x \in A^o \cup S\backslash\overline{A} \text{ and } x \in B^o \cup S\backslash\overline{B} \\
&x \notin \mathrm{bd}(A) \text{ and } x \notin \mathrm{bd}(B) \\
\therefore\ &x \notin \mathrm{bd}(A \cap B) \subset \mathrm{bd}(A)
\end{aligned}$$

22.3 Inward-pointing vectors at the boundary

Let M be a manifold with boundary and let $p \in \partial M$. Show that $X_p \in T_pM$ is inward-pointing if and only if in any coordinate chart $(U, x^1, \ldots, x^n)$ centered at p, the coefficient of $(\frac{\partial}{\partial x^n})_p$ in X_p is positive.

22.4 Smooth outward-pointing vector field along the boundary

Show that the vector field $X = \sum \rho_\alpha X_\alpha$ defined in the proof of Proposition 22.10 is a smooth outward-pointing vector field along ∂M.

22.5 Boundary orientation

Let M be an oriented manifold with boundary, ω an orientation form for M, and X a C^∞ outward-pointing vector field along ∂M.

(a) if τ is another orientation form on M, then $\tau = f\omega$ for a C^∞ everywhere-positive function f on M. Show that $\iota_X\tau = ft_X\omega$ and therefore, $\iota_X\tau \sim \iota_X\omega$ on ∂M. (Here '$\sim$' is the equivalence relation defined in Subsection 21.4.)

(b) Prove that if Y is another C^∞ outward-pointing vector field along ∂M, then $\iota_X\omega \sim \iota_y\omega$ on ∂M.

22.6 **Induced atlas on the boundary**

Assume $n \geq 2$ and let (U, ϕ) and (V, ψ) be two charts in an oriented atlas of an orientable n-manifold M with boundary. Prove that if $U \cap V \cap \partial M \neq 0$, then the restriction of the transition function $\psi \circ \phi^{-1}$ to the boundary $B := \phi(U \cap V) \cap \partial\mathcal{H}^n$,

$$(\psi \circ \phi^{-1})|_B : \phi(U \cap V) \cap \partial\mathcal{H}^n \to \psi(U \cap V) \cap \partial\mathcal{H}^n,$$

has positive Jacobian determinant. (*Hint:* Let $\phi = (x^1, \ldots, x^n)$ and $\psi = (y^1, \ldots, y^n)$. Show that the Jacobian matrix of $\phi \circ \phi^{-1}$ in local coordinates is block triangular with $J(\psi \circ \phi^{-1})|_B$ and $\frac{\partial y^n}{\partial x^n}$ as the diagonal blocks, and that $\frac{\partial y^n}{\partial x^n} > 0$.)

Thus, if $\{(U_\alpha, \phi_\alpha)\}$ is an oriented atlas for a manifold m with boundary, then the induced atlas $\{(U_\alpha \cap \partial M, \phi_\alpha|_{U_\alpha \cap \partial M})\}$ for ∂M is oriented.

22.7 **Boundary orientation of the left half-space**

let M be the left half-space

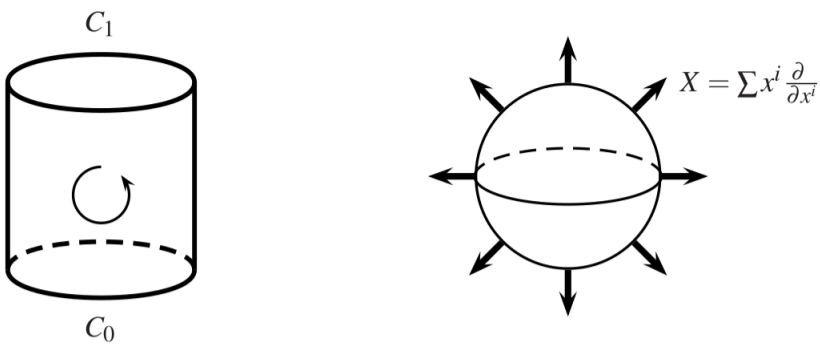
$$\{(x^1, \cdots, x^n) \in \mathbb{R}^n \mid x^1 \leq 0\},$$

with orientation from $dx^1 \wedge \cdots \wedge dx^n$. Show that an orientation from the boundary orientation on $\partial M = \{(0, x^2, \ldots, x^n) \in \mathbb{R}^n\}$ is $dx^2 \wedge \cdots \wedge dx^n$.

Unlike the upper half-space $\mathcal{H}^n$, whose boundary orientation takes on a sign (Example 22.13), this exercise shows that the bouddary orientation for the left half-space has no sign. For this reason some authors use the left half-space as the model of a manifold with boundary.

22.8 **Boundary orientation on a cylinder**

Let M be the cylinder $S^1 \times [0, 1]$ with the counterclockwise orientation when viewed from the exterior (Figure 22.7(a)). Describe the boundary orientation on $C_0 = S^1 \times \{0\}$ and $C_1 = s^1 \times \{1\}$.



(a) Oriented cylinder. (b) Radial vector field on a sphere.

Fig. 22.7. Boundary orientations.

22.9 **Boundary orientation on a sphere**

Orient the unit sphere S^n in $\mathbb{R}^{n+1}$ as the boundary of the closed unit ball. Show that an orientation form on S^n is

$$\omega = \sum_{i=1}^{n+1} (-12)^{i-1} x^i dx^1 \wedge \cdots \wedge \hat{dx} i \wedge \cdots \wedge dx^{x+1},$$

where the caret ˆ over dx^i indicates that dx^i is to be omitted. (*Hint:* An outward pointing vector field on S^n is the radial vector field $X = \sum x^i \frac{\partial}{\partial x^i}$ as in Figure 22.7(b).)

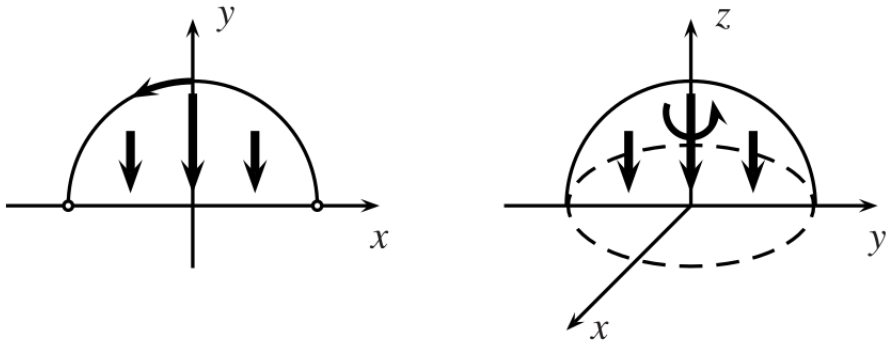


Fig. 22.8. Projection of the upper hemisphere to a disk.

22.10 **Orientation on the upper hemisphere of a sphere**

Orient the unit sphere S^n in $\mathbb{R}^{n-1}$ as the boundary of the closed unit ball. Let U be the upper hemisphere

$$U = \{x \in S^n \mid x^{n+1} > 0\}$$

It has a coordinate chart on the sphere with coordinates $x^1, \ldots, x^n$.

(a) Find an orientation form on U in terms of $dx^1, \ldots, dx^n$.

(b) Show that the projection map $\pi : U \to \mathbb{R}^n$.

$$\pi(x^1, \ldots, x^n, x^{n+1}) = (x^1, \ldots, x^n),$$

is orientation-preserving if and only if n is even (Figure 22.8).

22.11 **Antipodal map on a sphere and the orientability of $\mathbb{R}P^n$**

(a) the antipodal map $a : S^n \to S^n$ on the n-sphere is defined by

$$a(x^1, \ldots, x^n) = (-x^1, \ldots, -x^n)$$

Show that the antipodal map is orientation-preserving if and only if n is odd.
The differential is

$$da(x^i) = -I_{n+1}$$

restricting da to the tangent space and taking the determinant

$$da_x|_{T_x(S_n)} = -I_{n-1}$$
$$\det\left(da_x|_{T_x(S_n)}\right) = (-1)^{n-1}$$

which is orientation preserving when n is odd and orientation reversing when n is even.

(b) Use part (a) and Problem 21.6 to prove that an odd-dimensional real projective space $\mathbb{R}P^n$ is orientable.